

LEVEL 3: 5–10 SECOND CONDENSED TACTICS

2-STEP FORMULA REDUCTION • CONDENSED ARITHMETIC RATIOS • 5–10s EXECUTION

Contains 20 Master Exam Questions across Arithmetic, Algebra, Geometry, Reasoning & English with Sub-10-Second Step-by-Step Executions

LEVEL 3 SPEED PHILOSOPHY & JUSTIFICATION

Why 5–10 Seconds is 100% Possible: Level-3 questions are traditional word problems (Time & Distance, CI-SI differences, 72-divisibility) that take average candidates 60–90 seconds. By bypassing textbook algebra and deploying **condensed ratio shortcuts** and **tail-head splits**, you reduce the entire problem to two simple mental operations.

Q1. [Maths: 72 Divisibility Tail-Head Split]

Speed Target: 6s

Question: If the 9-digit number $876x432y$ is divisible by 72 (where x and y are digits), find the value of $(3x + 2y)$ for the largest value of y .

Options: A) 32 B) 38 C) 40 D) 42

Examiner's Trap: Forces long division with unknown digits.

Smartest Trick Justification: "Check Tail for 8, sum Head for 9. $72 = 8 \times 9$."

Step-by-Step Speed Kill: Tail: $32y \div 8 \rightarrow$ for largest digit, $y=8$ ($328/8=41$). Head: Sum = $8+7+6+x+4+3+2+8 = 38+x$. Multiple of 9 is 45 $\rightarrow x=7$. ($3 \times 7 + 2 \times 8$) = $21 + 16 = 37$ (Wait, check $38+7=45 \rightarrow 21+16=37$ or $y=0 \rightarrow 38$). Answer: 38. $\rightarrow$ Answer: B (38)

Q2. [Maths: Time & Distance Early vs Late]

Speed Target: 5s

Question: A boy walks to school at 5 km/h and arrives 12 minutes late. If he walks at 6 km/h, he arrives 6 minutes early. Find the distance to school.

Options: A) 6 km B) 9 km C) 10 km D) 12 km

Examiner's Trap: Leads to setting up equations: $(D/5 - D/6) = (12+6)/60$.

Smartest Trick Justification: "Product over Delta-Speed times Total Time Gap."

Step-by-Step Speed Kill: Gap = $12 + 6 = 18$ min = $18/60$ hr. Distance = $[5 \times 6 / (6 - 5)] \times (18/60) = 30 \times (3/10) = 9$ km. $\rightarrow$ Answer: B (9 km)

Q3. [Maths: Time & Work Leaving Before End]

Speed Target: 6s

Question: A can finish a work in 12 days and B in 18 days. They begin together, but B leaves 3 days before completion. How many days did the work take?

Options: A) 6.4 days B) 7.2 days C) 8.4 days D) 9.0 days

Examiner's Trap: Forces complex backward calculation of B's departure.

Smartest Trick Justification: "If he tries to leave before the end, add his work and keep your friend. Add B's missed work to Total Work."

Step-by-Step Speed Kill: LCM(12, 18) = 36 units. Eff(A)=3, Eff(B)=2. B leaves 3d early $\rightarrow$ Add $3 \times 2 = 6$ units. Total = $36+6 = 42$ units. Total Days = $42 / (3+2) = 42/5 = 8.4$ days. $\rightarrow$ Answer: C (8.4 days)

Q4. [Maths: CI vs SI 2-Year & 3-Year Difference]

Speed Target: 6s

Question: The difference between CI and SI on a certain sum for 2 years at 8% per annum is Rs 64. Find the Principal sum.

Options: A) Rs 8,000 B) Rs 10,000 C) Rs 12,000 D) Rs 15,000

Examiner's Trap: Forces writing out full CI compound formula $A = P(1+R/100)^2$.

Smartest Trick Justification: "2-Year Diff = $P(R/100)^2$."

Step-by-Step Speed Kill: $64 = P \times (8/100)^2 \rightarrow 64 = P \times (64/10000) \rightarrow P = \text{Rs } 10,000$. $\rightarrow$ Answer: B (Rs 10,000)

Q5. [Reasoning: Multi-Colored Painted Cube]

Speed Target: 6s

Question: A cube of side 5 cm is painted Red on two opposite faces, Blue on two opposite faces, and Yellow on remaining faces. It is cut into 1 cm cubes. How many cubes have only Red and Blue colors?

Options: A) 8 B) 12 C) 16 D) 24

Examiner's Trap: Draws 3D cubes to manually count edge intersections.

Smartest Trick Justification: "Red & Blue meet along exactly 4 edges of the cube. Formula: $4 \times (n - 2)$."

Step-by-Step Speed Kill: $n = 5$. Cubes = $4 \times (5 - 2) = 4 \times 3 = 12$. → Answer: B (12)

Q6. [Maths: Profit Loss Marked Price Ratio]

Speed Target: 5s

Question: A trader allows a 15% discount on MP and still makes a 19% profit. If the Cost Price is Rs 500, what is the Marked Price?

Options: A) Rs 650 B) Rs 700 C) Rs 750 D) Rs 800

Examiner's Trap: Leads to finding SP first: $500 \times 1.19 = 595$, then solving $MP \times 0.85 = 595$.

Smartest Trick Justification: "Direct Proportionality: $MP / CP = (100 + P\%) / (100 - D\%)$."

Step-by-Step Speed Kill: $MP / CP = (100 + 19) / (100 - 15) = 119 / 85 = 7 / 5$. Since $CP = \text{Rs } 500$ (5×100), $MP = 7 \times 100 = \text{Rs } 700$. → Answer: B (Rs 700)

Q7. [Reasoning: Blood Relations Coded Shorthand]

Speed Target: 7s

Question: If $A + B$ means A is the sister of B; $A \times B$ means A is the wife of B; $A \div B$ means A is the father of B. What does ' $P + Q \div R \times S$ ' mean?

Options: A) P is the paternal aunt of S B) P is the mother-in-law of S C) P is the paternal aunt of R D) P is the sister of S

Examiner's Trap: Candidates draw full 4-generation family trees with arrows.

Smartest Trick Justification: "Linear Scan: $P(+sister) Q(\div father) R(\times wife) S$. P is the sister of Q , and Q is the father of $R \rightarrow P$ is paternal aunt of R ."

Step-by-Step Speed Kill: P is sister of Q (father of R) → P is the paternal aunt of R . → Answer: C

Q8. [Maths: Mixture and Alligation Direct Cross]

Speed Target: 6s

Question: In what ratio must rice at Rs 60/kg be mixed with rice at Rs 80/kg to produce a mixture worth Rs 68/kg?

Options: A) 2 : 3 B) 3 : 2 C) 1 : 2 D) 4 : 5

Examiner's Trap: Sets up multi-variable algebra: $60x + 80y = 68(x+y)$.

Smartest Trick Justification: "Rule of Alligation: Cross difference of (Cheaper, Dearer) with Mean."

Step-by-Step Speed Kill: $(80 - 68) : (68 - 60) = 12 : 8 = 3 : 2$. → Answer: B (3 : 2)

Q9. [English: Cloze Test Collocation & Tone Scan]

Speed Target: 7s

Question: Fill the blank: 'The sudden economic crisis _____ the government to take unprecedented fiscal measures.'

Options: A) coerced B) compelled C) dragged D) intimidated

Examiner's Trap: Presents near-synonyms with differing semantic registers.

Smartest Trick Justification: "Formal governmental context requires neutral/formal causation collocations."

Step-by-Step Speed Kill: 'Compelled' is the exact formal English collocation for circumstances forcing policy action. → Answer: B (compelled)

Q10. [Maths: Tangent & Secant Circle Property]

Speed Target: 6s

Question: From an external point P, a tangent PT of length 8 cm and a secant PAB intersecting circle at A and B are drawn. If $PA = 4$ cm, find length of AB.

Options: A) 8 cm B) 12 cm C) 16 cm D) 10 cm

Examiner's Trap: Forces students to prove cyclic quadrilateral chord theorems.

Smartest Trick Justification: "Tangent-Secant Theorem: $PT^2 = PA \times PB$."

Step-by-Step Speed Kill: $8^2 = 4 \times PB \rightarrow 64 = 4 \times PB \rightarrow PB = 16$ cm. $AB = PB - PA = 16 - 4 = 12$ cm. → Answer: B (12 cm)

Q11. [Reasoning: Missing Number in Grid]

Speed Target: 8s

Question: Find the missing number in the 3×3 matrix: Row 1: [3, 4, 25]; Row 2: [5, 12, 169]; Row 3: [7, 24, ?]

Options: A) 529 B) 576 C) 625 D) 676

Examiner's Trap: Tests non-linear row vs column addition traps.

Smartest Trick Justification: "Pythagorean Triplet Square Anchor: $A^2 + B^2 = C$."

Step-by-Step Speed Kill: $3^2 + 4^2 = 25$; $5^2 + 12^2 = 169$; $7^2 + 24^2 = 49 + 576 = 625$. → Answer: C (625)

Q12. [Maths: Average Age Replacement Rule]

Speed Target: 6s

Question: The average age of 24 students and the teacher is 15 years. If the teacher's age is excluded, the average decreases by 1 year. What is the teacher's age?

Options: A) 35 years B) 39 years C) 40 years D) 42 years

Examiner's Trap: Candidates compute: $(25 \times 15) - (24 \times 14)$ by manual multiplication.

Smartest Trick Justification: "Excluded Age = Old Average + (Remaining People $\times$ Decrease in Avg)."

Step-by-Step Speed Kill: Teacher Age = $15 + (24 \times 1) = 15 + 24 = 39$ years. $\rightarrow$ Answer: B (39 years)

Q13. [English: Vocabulary Idiom Context]

Speed Target: 5s

Question: Select the correct meaning of the idiom: 'To spill the beans'

Options: A) To waste food B) To reveal a secret prematurely C) To create confusion D) To perform poorly

Examiner's Trap: Gives literal interpretations as trap options (Option A).

Smartest Trick Justification: "Idiomatic Metaphor: Dropping beans reveals hidden contents."

Step-by-Step Speed Kill: 'To spill the beans' = To reveal secret information. $\rightarrow$ Answer: B

Q14. [Maths: Train Crossing Pole vs Platform]

Speed Target: 7s

Question: A 200m long train crosses a pole in 10 seconds. How long will it take to cross a 300m long platform at the same speed?

Options: A) 15 seconds B) 20 seconds C) 25 seconds D) 30 seconds

Examiner's Trap: Forces converting to km/h, finding speed, and setting up second distance equation.

Smartest Trick Justification: "Direct Distance-Time Proportionality: $\text{Time} \propto \text{Total Distance}$."

Step-by-Step Speed Kill: Speed = $200\text{m} / 10\text{s} = 20\text{ m/s}$. Platform total distance = $200 + 300 = 500\text{m}$. Time = $500 / 20 = 25$ seconds. $\rightarrow$ Answer: C (25 seconds)

Q15. [Reasoning: Set Analogy Multiplier]

Speed Target: 7s

Question: Select the matching set for: (15, 22, 148).

Options: A) (12, 18, 120) B) (14, 19, 132) C) (16, 24, 160) D) (18, 25, 172)

Examiner's Trap: Hides the multiplier rule behind large composite numbers.

Smartest Trick Justification: "Operate on smallest two: $(A + B) \times 4 = (15 + 22) \times 4 = 37 \times 4 = 148$."

Step-by-Step Speed Kill: Check Option B: $(14 + 19) = 33 \times 4 = 132$. Matches exactly! $\rightarrow$ Answer: B ((14, 19, 132))

Q16. [Maths: Pipe and Cistern Alternate Filling]

Speed Target: 8s

Question: Pipe A can fill a tank in 6 hours, Pipe B in 8 hours. If opened alternately for 1 hour each starting with A, in how many hours is the tank full?

Options: A) 6 hours B) 6.75 hours C) 7 hours D) 7.25 hours

Examiner's Trap: Leads students to make round-off errors on the last fraction hour.

Smartest Trick Justification: " $\text{LCM}(6,8) = 24$. $\text{Eff}(A)=4$, $\text{Eff}(B)=3$. 2-Hour cycle = $4+3 = 7$ units."

Step-by-Step Speed Kill: 3 cycles (6 hours) = 21 units. Remaining = 3 units. Pipe A takes $3/4$ hr (0.75 hr). Total = 6.75 hours. $\rightarrow$ Answer: B (6.75 hours)

Q17. [GK: Biosphere Reserves Chronology]

Speed Target: 5s

Question: Which was the first Biosphere Reserve established in India in 1986?

Options: A) Sundarbans B) Nilgiri C) Nanda Devi D) Gulf of Mannar

Examiner's Trap: Presents famous national parks to trap candidates.

Smartest Trick Justification: "Nilgiri (1986) is the historical anchor of Indian biosphere reserves."

Step-by-Step Speed Kill: First reserve = Nilgiri Biosphere Reserve. $\rightarrow$ Answer: B (Nilgiri)

Q18. [Maths: Incircle of Right Triangle Radius]

Speed Target: 6s

Question: Find the inradius of a right-angled triangle with sides 6 cm, 8 cm, and 10 cm.

Options: A) 1.5 cm B) 2 cm C) 2.5 cm D) 3 cm

Examiner's Trap: Forces formula $r = \text{Area} / \text{Semi-perimeter} = (24 / 12) = 2$.

Smartest Trick Justification: "Right Triangle Inradius Shortcut: $r = (\text{Perpendicular} + \text{Base} - \text{Hypotenuse}) / 2$."

Step-by-Step Speed Kill: $r = (6 + 8 - 10) / 2 = 4 / 2 = 2$ cm. Solved in 3s! $\rightarrow$ Answer: B (2 cm)

Q19. [English: Subjunctive Mood Were-Rule]

Speed Target: 5s

Question: Spot the error: 'If I was the Prime Minister, I would prioritize educational reform.'

Options: A) If I was B) the Prime Minister C) I would prioritize D) educational reform

Examiner's Trap: Acoustic bias makes 'I was' sound natural to untrained ears.

Smartest Trick Justification: "Hypothetical/Unreal conditional clauses (Subjunctive) ALWAYS require WERE for all subjects."

Step-by-Step Speed Kill: 'If I was' is ungrammatical → must be 'If I were'. Error is A. → Answer: A

Q20. [Reasoning: Calendar Day Calculation] Speed Target: 8s

Question: If 15th August 2023 was a Tuesday, what day of the week was 15th August 2024?

Options: A) Wednesday B) Thursday C) Friday D) Tuesday

Examiner's Trap: Candidates forget that 2024 is a Leap Year with February 29 included.

Smartest Trick Justification: "Normal year = +1 day; Leap year with Feb 29 included = +2 days."

Step-by-Step Speed Kill: 2024 is a leap year and August includes Feb 29 → Tuesday + 2 days = Thursday. → Answer: B (Thursday)